

Math Essentials for ML

The 3 math areas you need to master Machine Learning

1. Probability & Statistics (50% of ML math)

Used in almost every ML model. Focus on: random variables, probability distributions, Bayes theorem, expectation & variance, Central Limit Theorem.

Resources:

- StatQuest with Josh Starmer — <https://www.youtube.com/@statquest>
- Harvard Probability (free) — <http://probabilitybook.net>

Neural networks = matrix operations. Focus on: vectors & matrices, matrix multiplication, eigenvalues, projections.

Resources:

- 3Blue1Brown: Essence of Linear Algebra — https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
- Khan Academy Linear Algebra — <https://www.khanacademy.org/math/linear-algebra>

3. Calculus (15% of ML math)

Used for training ML models. Focus on: derivatives, gradients, gradient descent.

Resources:

- Mathematics for Machine Learning (free book) — <https://mml-book.github.io>
- Khan Academy Calculus — <https://www.khanacademy.org/math/calculus-1>

Bonus: See ML Math in Action

Best series to see ML math applied in real code:

- Andrej Karpathy: Neural Networks Zero to Hero — <https://www.youtube.com/playlist?list=PLAqhlrjkbxWl23v9cThsA9GvCAUhRvKZ>